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# SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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## ABSTRACT

Mixture of Experts (MoE) models are now a central scaling mechanism for large language models, but their representational contribution relative to dense networks remains mechanistically unclear. This paper makes that contribution explicit by extending toy-model analyses of *superposition* to MoEs and asking which sparsity notion best explains their feature geometry. We report three findings. First, the dense-model drivers of phase changes—feature sparsity and feature importance—do not by themselves explain MoE behavior; network sparsity, the ratio of active to total experts, is the more salient organizing variable. Second, we introduce expert-level measures of superposition and show that increasing network sparsity produces more *monosemantic* feature representations, even under comparable parameter budgets. Third, we reframe expert specialization as monosemantic feature representation rather than load balance alone, and show that appropriately initialized experts organize around coherent feature combinations. Together, these results identify network sparsity as a mechanism by which MoEs can make representations more interpretable while maintaining comparable reconstruction performance in our toy setting.

## 1 INTRODUCTION

Mixture of Experts (MoEs) are widely used in state-of-the-art language models such as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), where their headline contribution is efficient scaling and improved performance (Shazeer et al., 2017; Fedus et al., 2022). Work on Expert-Choice routing and ST-MoE further improves routing, stability, and transfer (Zhou et al., 2022; Zoph et al., 2022). Other theoretical and empirical analyses show that learnable routers can recover latent cluster structure, giving one account of why experts specialize (Dikkala et al., 2023). What remains missing, and what this paper targets, is a mechanistic account of how MoEs represent features differently from dense networks.

Recent interpretability-oriented MoE systems make expert behavior more transparent by changing the architecture. Yang et al. (2025b) propose MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling expert counts to support capability editing through expert activation. These works motivate our study but leave a narrower contribution unfilled: isolating how ordinary MoE routing changes feature representation, superposition, and specialization without relying on additional interpretability-specific design. This gap aligns with Mu & Lin (2025), who identify mechanistic interpretability as an open challenge for MoE research.

The central obstacle we study is superposition: the ability of a model to represent more features than it has dimensions. Superposition is valuable for capacity because sparse features can be packed into shared neurons, but it reduces interpretability because individual neurons become polysemantic rather than cleanly feature-aligned.

MoE architectures add a distinct explanatory variable to this problem: network sparsity. Dense models activate the same parameters for every input, whereas MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). This difference suggests the

paper’s main hypothesis: dense models exploit feature sparsity by packing sparse features into shared dimensions, while MoEs can use conditional computation to allocate experts to particular feature combinations and thereby reduce global interference.

We therefore organize the paper around three reviewer-facing questions: (1) do MoEs exhibit less superposition than dense models with comparable parameter budgets; (2) do MoE experts show the same discrete phase changes that dense toy models show when feature sparsity and relative importance vary; and (3) can expert specialization be defined by represented features rather than by load-balancing statistics alone?

Using simple models that extend Elhage et al. (2022)’s framework to MoEs, we make three contributions. **First**, we show that MoEs do not inherit the sharp dense-model phase changes; as network sparsity increases, transitions between ignored, superposed, and monosemantic representations become more continuous. **Second**, we introduce MoE-compatible measurements of feature dimensionality and features per dimension, and use them to show that MoEs exhibit greater monosemanticity—that is, less superposition—than dense models with comparable active and total parameter budgets. **Third**, we propose an interpretability-focused definition of expert specialization based on monosemantic feature representation and show that, under appropriate initialization, experts organize around coherent feature combinations rather than arbitrary load balance.

## 2 RELATED WORK

**Superposition and Feature Representations.** The Linear Representation Hypothesis frames concepts as directions in activation space (Park et al., 2024), but interpretable features can outnumber available dimensions. Elhage et al. (2022) formalized this mismatch as superposition and showed that dense models increase capacity by packing features into non-orthogonal directions, at the cost of polysemanticity. Subsequent work emphasizes both the efficiency of this packing (Scherlis et al., 2025) and the need for post-hoc tools such as Sparse Autoencoders to recover monosemantic features (Bricken et al., 2023). Other studies examine how correlations shape superposition (Prieto et al., 2025) and how interference patterns arise or can be mitigated (Gurnee et al., 2023). Our contribution is to move this feature-geometry analysis from dense models to MoEs and to show that network sparsity, not feature sparsity alone, is a key driver of the resulting representation strategy.

**Interpretability of MoEs.** Although MoEs are a standard scaling architecture for large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), their internal representations are less mechanistically understood than those of dense models. Existing analyses often emphasize system-level behavior: routing stability (Zoph et al., 2022), expert choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). Recent interpretability-driven architectures encourage specialization through predefined concept routing (Yang et al., 2025b) or expert counts matched to vocabulary sizes (Park et al., 2025). We take a complementary route: rather than forcing specialization, we measure whether specialization emerges in feature geometry, showing that MoEs can reduce superposition and motivating a feature-based definition of expert specialization.

## 3 BACKGROUND

Mechanistic interpretability often proceeds by reverse engineering a model’s representation into human-interpretable concepts, or ‘features’. The framing we use throughout the paper assumes these features are approximately linear: any hidden state  $h$  can be described as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

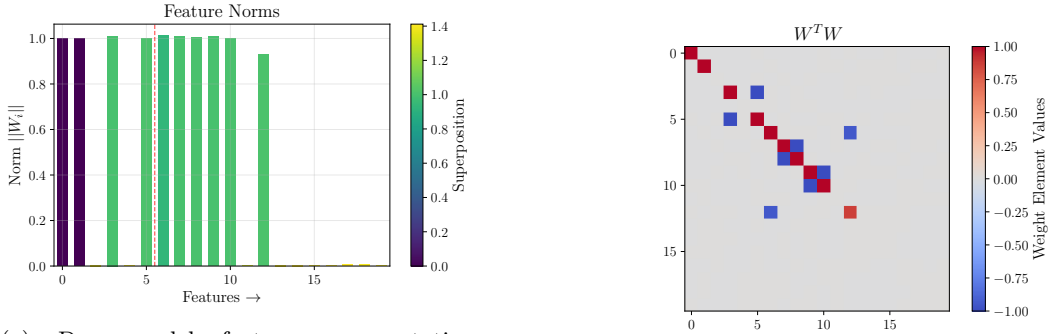
where  $\vec{f}_i$  denotes the direction for feature  $i$ ,  $\alpha_i$  denotes that feature’s activation strength, and  $F$  is the full set of represented features. This notation makes the paper’s contribution measurable: we can ask whether MoE experts represent these directions cleanly or pack them together in superposition.

The *superposition hypothesis* states that models can represent many more features than dimensions, i.e.  $|F| > m$  for  $h \in \mathbb{R}^m$  (Elhage et al., 2022). Achieving this requires feature vectors in  $F$  to be packed into non-orthogonal directions. When two features interfere,  $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$ , we say they are in *superposition*. This packing can be useful when features are sparse, a common property of online text, but it directly creates the interpretability problem we measure:  $\alpha_i$  may contain activation caused by other features rather than by feature  $i$  alone.

Monosemanticity is usually a neuron-level property: a neuron’s activation corresponds cleanly to a single  $\alpha_i$ , so the feature is basis-aligned. Orthogonal features that are not basis-aligned can still make individual neurons *polysemantic*, even when feature interference is low. To keep our contribution focused and measurable, we study reductions in superposition rather than strict basis alignment. Thus, when we call a model, expert, or feature “more monosemantic,” we mean that it exhibits less superposition in this feature-interference sense.

## 4 DEMONSTRATING SUPERPOSITION

Our first empirical contribution tests a common intuition: an MoE might simply be a collection of dense models, with each expert learning dense-like feature geometry on its routed inputs. We evaluate that intuition directly by comparing dense and MoE representations through the amount and structure of superposition.



(a) Dense-model feature representation strength, measured by each feature weight-vector norm  $\|W_i\|$ ; color highlights whether the represented feature is in superposition (green) or comparatively monosemantic (purple).

(b) Dense-model interference map, where each  $W^T W$  entry gives  $(\hat{W}_i \cdot W_j)$  and makes pairwise feature interference visible.

Figure 1: Baseline dense-model feature geometry for  $n = 20$  features and  $m = 6$  hidden dimensions, with importance  $I = 0.7^i$  and uniform feature density  $(1 - S) = 0.1$ . The figure establishes the comparison target: feature norms show what is learned, while color reports superposition via  $\sum_j (\hat{W}_i \cdot W_j)^2$ .

### 4.1 EXPERIMENTAL SETUP

The experimental setup is designed to isolate the representational contribution of MoE routing. We extend Elhage et al. (2022)’s reconstruction task by asking how a MoE projects a high-dimensional vector  $x \in \mathbb{R}^n$  into lower-dimensional expert states  $h \in \mathbb{R}^m$  and then reconstructs the original input.

**The input distribution.** The input vector  $x$  represents activations from an idealized disentangled model: each coordinate  $x_i$  is a distinct, independent feature, so the input space begins as perfectly neuron-aligned and monosemantic. We sample  $x_i \sim U(0, 1)$  and set  $P(x_i = 0) = S$  for sparsity  $S \in [0, 1)$ . To keep feature utility explicit, each feature  $x_i$  receives an importance weight  $I_i$  in the reconstruction loss. For phase-change experiments, we isolate the last feature by scaling its magnitude with  $r \in \mathbb{R}^+$ , so  $I = (1, 1, \dots, r)$ . These

controls let us separate the effects of feature sparsity, feature importance, and network sparsity on expert representations.

**Model Architecture.** The MoE contains  $E$  experts, each parameterized by a weight matrix  $W^e \in \mathbb{R}^{m \times n}$  and bias  $b^e \in \mathbb{R}^n$ . A learned router  $g(x) = \text{softmax}(W^r x)$  with  $W^r \in \mathbb{R}^{E \times n}$  assigns each input to the top- $k$  experts. Each active expert computes a lower-dimensional state  $h^e = W^e x$  and reconstruction  $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ <sup>1</sup>. The MoE output is the weighted sum over active experts,  $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ , where  $w_e$  are the renormalized gating weights. This architecture lets us attribute changes in superposition to conditional expert selection rather than to a different reconstruction objective.

Training uses an L2 reconstruction loss weighted by feature importance,  $I_i$ , with  $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$ . In Section 4, we add the standard auxiliary load-balancing loss (Fedus et al., 2022),  $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$ , to avoid expert-collapse local minima in which some experts are ignored. Here,  $N$  is the number of experts,  $f_e$  is the fraction of batch samples routed to expert  $e$ ,  $P_e$  is that expert’s average gating probability across the batch, and  $\alpha = 0.01$  sets the penalty strength. In Section 4, we omit this auxiliary term so the measured superposition reflects the MoE architecture’s intrinsic bias rather than an explicit balancing objective.

## 4.2 MEASURING FEATURE CAPACITY

To compare feature geometry across architectures, we measure two quantities that directly expose the paper’s contribution: *representation strength* and *interference*. For expert  $e$ , the norm  $\|W_i^e\|$  measures how strongly feature  $i$  is represented:  $\|W_i^e\| \approx 1$  indicates full representation, while a norm near zero indicates that the feature is not learned. We then use the Gram matrix  $W^\top W$  to visualize interference, with off-diagonal entries showing when one represented feature affects another.

Figures 1a and 2a show that the dense model and MoE learn a comparable number of features (10 vs 8) with similar norms under equal total parameters ( $m_{\text{dense}} = \sum m_{\text{experts}} = 6$ ). The salient difference is not how many features are represented, but where interference can occur. MoE experts still show local superposition, such as Expert 0 in Figure 2b, but the global structure is partitioned: unlike the dense matrix in Figure 1b, where any feature can interfere with any other, the MoE produces a block-diagonal pattern in which features routed to different experts have zero interference. This supports the first contribution: MoEs reduce the global scope of superposition by partitioning feature space across experts.

**Expert Feature Dimensionality.** To make this comparison quantitative, we adapt feature dimensionality to the expert level. Dimensionality measures the “fraction of a dimension” assigned to a feature (Elhage et al., 2022), so it directly captures whether a feature is represented monosemantically or shares capacity with others. For feature  $i$  in expert  $e$ , we define

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

$D_i^e$  ranges from zero, when the feature is not learned, to one, when the feature is represented monosemantically. Summing over experts gives the MoE’s total measured representational capacity,  $D = \sum_e \sum_{i=1}^n D_i^e$ .

**Efficient Packing.** A model packs features efficiently when the summed feature dimensionalities match its available embedding dimensions, i.e.  $\sum_{i=1}^n D_i^e \approx m$  (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For MoEs, the analogous relation is

<sup>1</sup>The output ReLU enforces non-negative reconstructions, matching the input distribution where  $x_i \in [0, 1]$  when nonzero. The off-diagonal terms in  $(W^e)^\top W^e$  create negative interference, which ReLU suppresses. When features are sparse, negative interference is effectively “free” because it is filtered to zero, incentivizing configurations with negative off-diagonal terms, such as antipodal pairs.

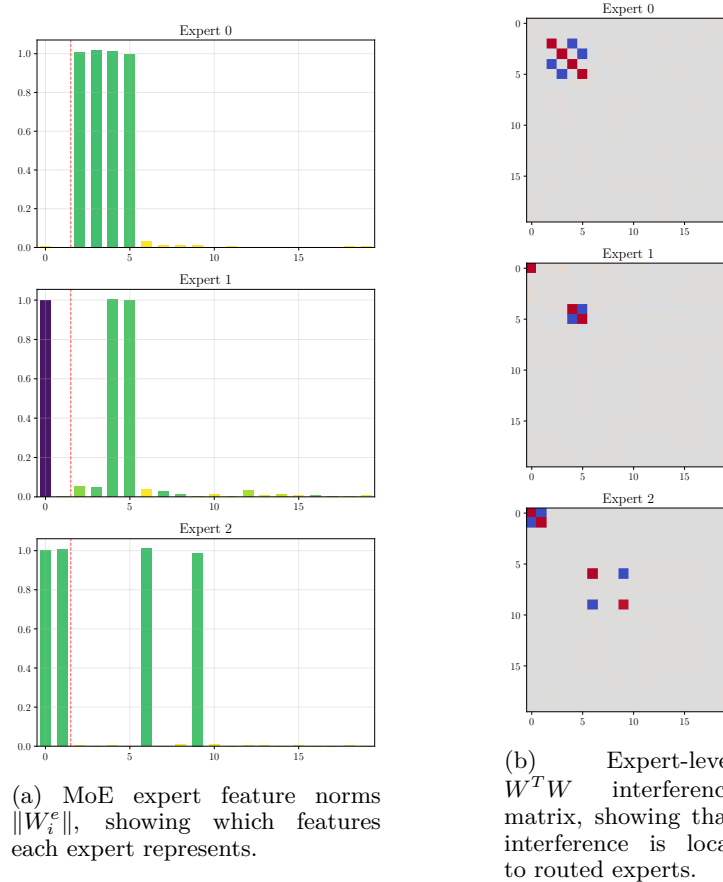


Figure 2: MoE feature geometry for  $n = 20$  features, 3 total experts, and  $m = 2$  hidden dimensions per expert under top- $k = 1$  routing, with importance  $I = 0.7^i$  and feature density  $1 - S = 0.1$ . Compared with Figure 1, the MoE contribution is a partitioned interference structure across experts.

$\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$ . Empirically, both dense and MoE models satisfy this dimensionality constraint, so the central difference is not packing efficiency. Instead, MoEs achieve similar capacity while redistributing interference across experts.

**Features per Dimension & Network Sparsity.** Because dense and MoE models both pack efficiently, features per dimension gives a sharper measure of the contribution we care about: how much superposition remains after routing partitions the input space. Values above one indicate superposition because the model represents more features than available dimensions. For a MoE, we define features per dimension as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where  $\|W\|_F^2$  is the Frobenius norm and  $p_e$  is the expected usage probability of expert  $e$  across a batch, computed as the average renormalized gating weight after top- $k$  routing.

Under *equal* total parameter counts, Figure 3 shows that the dense model has more features per dimension and therefore more superposition. The dense model uses superposition to learn more features than its total dimensions ( $n_{\text{learned}} > m_{\text{total}}$ ), while the MoE tends to remain near the monosemantic limit ( $n_{\text{learned}} \approx m_{\text{total}}$ ). Holding total parameters and the ratio  $k/E$  roughly fixed, increasing the number of experts further decreases features per dimension. This is the paper’s clearest network-sparsity result: *the greater the number of*

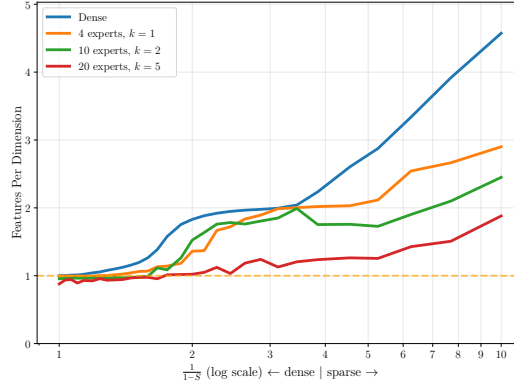


Figure 3: Features per dimension versus inverse feature density ( $\frac{1}{1-g}$ ) for **dense** and **MoE** architectures with uniform feature importance ( $I_i = 1.0$ ). This figure summarizes the superposition contribution: the dense model ( $n = 100$ ,  $m = 20$ ) shows the most features per dimension, while MoE superposition decreases as expert count increases: 4 experts with  $m = 5$ ,  $k = 1$  (orange); 10 experts with  $m = 2$ ,  $k = 2$  (green); and 20 experts with  $m = 1$ ,  $k = 5$  (red). All models have equal total parameters and similar  $k/E$ ; the dashed line at 1.0 marks monosemantic representation.

*experts, the less superposition in the model.* The dense model’s additional superposition yields consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with differences of  $\sim 0.03 - 0.08$  at a given sparsity. However, as expert count increases, MoE losses become consistently comparable to dense losses, suggesting that MoEs trade some dense-model packing for more monosemantic representations. Appendix A.2 gives theoretical intuition for this trade-off.

## 5 PHASE CHANGE

The second contribution asks whether MoEs follow the same phase-change story as dense toy models. The previous section showed that MoEs can learn a similar number of features while distributing interference across experts. Here we test whether input-distribution variables—feature sparsity and feature importance—still induce discrete, physics-inspired *phase changes*, or whether network sparsity changes the representational regime.

In this toy setting, each feature has three visible representational outcomes: it can be ignored, represented in superposition, or represented monosemantically. A phase change occurs when the boundary between these outcomes is sharp as feature sparsity or relative importance varies.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). Varying sparsity and relative importance can switch a dense model between ignoring a feature, superimposing it, or representing it cleanly; for example, greater feature sparsity can make superposition more attractive. Our contribution is to draw the same phase diagrams expert-by-expert for MoEs and show that these diagrams do not simply replicate the dense pattern.

**Setup.** We use the setup from Section 4.1, adding load-balancing loss only where needed to avoid specialization collapse, i.e. local minima in which the router ignores some experts entirely. We study three top- $k = 1$  architectures: (A)  $n=2$ ,  $m=1$ ; (B)  $n=3$ ,  $m=1$ ; and (C)  $n=3$ ,  $m=2$ . For each architecture, we plot expert-specific phase diagrams over feature sparsity and last-feature relative importance, and vary network sparsity by increasing the number of experts ( $E$ ) up to the number of input feature dimensions ( $n$ ).

For this phase-change contribution, we fix active rather than total parameters, so  $m_{dense} = km_{moe}$  up to router parameters. This choice isolates changes due to active capacity within an

architecture; otherwise, apparent phase differences could be explained by total architectural capacity. One comparison should therefore be read carefully: 4.C.1/1 has the same number of active parameters as 4.B.X/2, but the B models encode the same  $n = 3$  input features with only one hidden dimension ( $m = 1$ ), whereas C uses two hidden dimensions ( $m = 2$ ). This distinction matters when interpreting specialization through superposition.

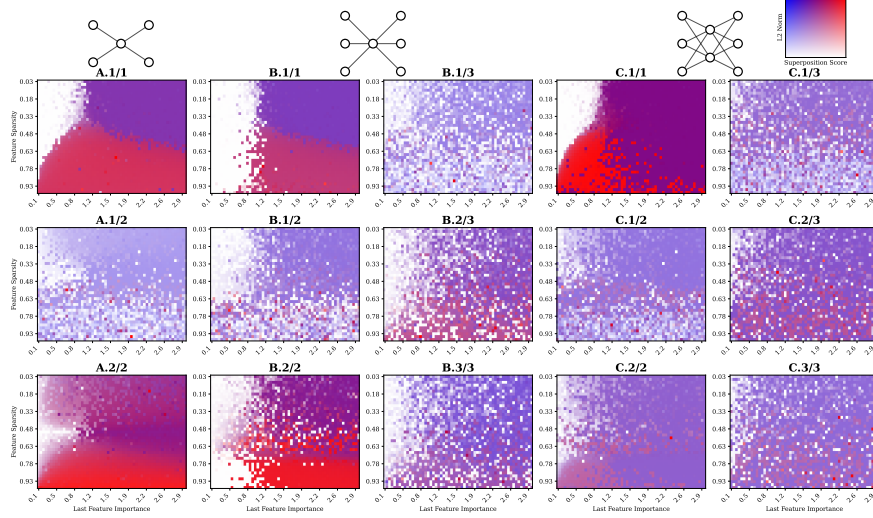


Figure 4: Expert-level phase diagrams for the last input feature. Each cell decodes whether a particular expert ignores the feature (white), represents it monosemantically (blue-purple), or represents it in superposition (red). We combine feature norm ( $\|W_n\|^2$ ) with superposition score ( $\sum_{j < n} (\tilde{W}_n \cdot W_j)^2$ ) while varying feature sparsity  $S \in [0.1, 1]$  and last-feature relative importance  $I_n \in [0.1, 3]$ , where  $n$  denotes the last of  $n$  total features. For each cell, we train ten models with load-balancing loss and report the lowest-loss run. Cell  $(i, j)$  in subfigure X.e/E denotes expert  $e$  of  $E$  total experts for architecture X at  $I_n = i$  and  $S = j$ ; X.1/1 is the dense baseline.

**Results and Takeaways.** The dense, single-expert baselines show clear phase changes in every architecture (Figure 4.X.1/1), reproducing the qualitative behavior of Elhage et al. (2022). The contribution of adding experts is that these sharp boundaries disappear. Some  $E = 2$  experts still resemble the corresponding dense phase diagrams (Figure 4.X.2/2), but their transitions are more continuous. In each setting, one expert becomes more monosemantic and specializes in the relatively important feature. Experts that diverge from the dense pattern have uniformly lower superposition scores, visible as bluer regions. This reinforces the previous section’s result: MoEs favor lower-superposition, more monosemantic representations than dense counterparts.

The architecture-specific patterns clarify what network sparsity changes. In the  $n = 2, m = 1$  setup (Figure 4.A), the dense model ignores the last feature when feature sparsity is low, while the comparable MoE preserves that feature more often because routing gives it usable capacity. With three input dimensions (Figure 4.B), the MoEs cannot show the same benefit because the experts already superimpose the other two features, leaving no room for the third feature in one hidden dimension. In architecture B, even high-sparsity superposition is insufficient to represent all features, yet the sharp dense-style phase change is still mostly absent, except for the mixed but partially discrete 0.5–0.7 sparsity region in 4.B.2/2. For  $m = 2$ , the dense model’s white region (Figure 4.C.1/1) shows that the last feature is ignored when sparsity is mid-to-low and the feature is relatively less important. As network sparsity increases across Figure 4.C, the models represent the last feature with larger L2 magnitude ( $\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$ ), matching the dimensionality increase shown in Figure 3.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2, the sparsity interval from roughly 0.48 to 0.7 contains a heterogeneous mix of polysemanticity, monosemanticity, and feature ignorance. This middle region suggests that MoEs can have comparable loss under multiple represen-

tation strategies, making them less likely to commit to the clean low- and high-sparsity regimes seen in dense models. The absence of this pattern in dense baselines is further evidence that MoE routing creates distinct representational strategies.

**Conclusion.** These experiments all use  $\text{top-}k = 1$ , so only one expert is active for each input. Even under this restrictive setting, behavior changes substantially as soon as  $E = 2$ , including cases where hidden-dimensional capacity with superposition is sufficient to represent all features. The phase-change contribution is therefore negative but important: MoEs should not be interpreted as simple aggregations of dense models. Router-induced restriction to subsets of the feature domain changes what experts learn and how sharply they transition between representational strategies.

## 6 EXPERT SPECIALIZATION

The third contribution turns the reduced-superposition result into a definition of expert specialization. If MoEs make features more monosemantic, the next question is whether those monosemantic features are organized coherently within experts.

MoE specialization is often evaluated through load balancing across experts and inputs (Chaudhari et al., 2025). Load balance is useful for training and utilization, but it is not a mechanistic definition of specialization: an expert is specialized when it is used for the concepts it represents, not merely when it receives an average share of tokens.

We therefore define an expert as specialized when two conditions align: it *occupies* particular feature directions in input space, and it represents those same features relatively monosemantically. The contribution of this section is to show that these two conditions are correlated in our toy MoEs and that their alignment can also support balanced expert usage.

Because we fix  $k = 1$ , the router partitions feature space into convex cone regions (Appendix A.3), each assigned to one expert. Formally, if  $C$  is a cone and  $s$  is any positive scalar, then  $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$ . Thus, if a feature direction  $\vec{x}$  routes to an expert, every scaled version  $s\vec{x}$  routes to that same expert. We say that feature  $x$  is contained within expert  $e$ , or equivalently that expert  $e$  *occupies*  $x$ .

Appendix A.4 shows an initial empirical pattern: in small models, lower loss is associated with distributing the input space across more experts. That observation motivates the specialization test here. If an expert is assigned a region of input space, does it also represent the corresponding features more monosemantically? Our proposed definition predicts this link, and we test it in larger toy models, including  $m = 10$ .

For models with  $n > 2$ , we evaluate the specialization definition in both directions. First, we ask whether initializing experts to occupy particular input-space features makes those experts more monosemantic with respect to those features. Second, for features that an expert represents monosemantically, we ask whether the router also assigns those feature directions to the same expert.

Figure 5 makes the initialization-to-specialization link visible. With diagonal gate initialization, each expert monosemantically represents the single feature it initially occupied, and no other feature (Figure 5a). With ordered  $k$ -hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized on less important features, do not monosemantically represent those features or the five most important features they were not assigned. When feature importance order is broken by random  $k$ -hot initialization, each expert monosemantically represents only the most important feature or features among those it initially occupied (Figure 5c).

These results support the section’s main contribution: routed feature occupancy predicts monosemantic feature representation, but importance determines which occupied features receive clean representation. The pattern holds whether an expert is initialized with one important feature directly or with a set of features from which it selects the most important feature to represent monosemantically.

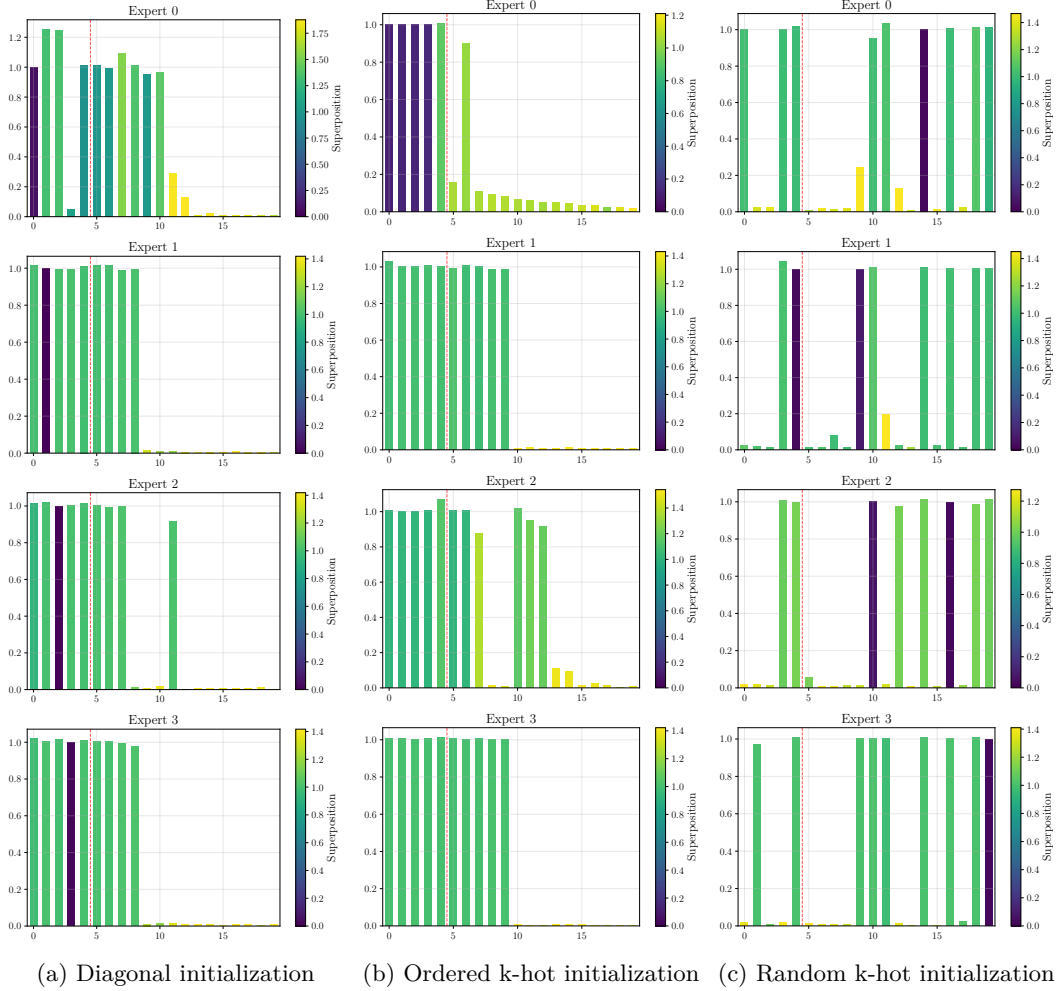


Figure 5: Specialization under three router initialization schemes, visualized with expert feature norms  $\|W_i^{(e)}\|$  and superposition color for  $n = 20, m = 5, E = 4, S = 0.1$ . In (a), the gate matrix is initialized on the main diagonal ( $W_i^T = \hat{e}_i$ , the basis vector for that dimension), while relative feature importance decreases exponentially from feature one to 20. In (b), the gate matrix uses an “ordered k-hot” initialization: the first expert aligns with the first five features, and each later expert aligns with the next five features; relative importance matches (a). In (c), the gate matrix uses a “random k-hot” initialization in which each expert receives five random, non-overlapping features that collectively cover all 20 features; relative importance still decreases exponentially but is randomly distributed across features.

Uniform feature importance weakens absolute monosemanticity: experts place all features into higher-interference superposition. Even in this harder setting, initialization remains visible in the feature geometry. The features an expert initially occupies are *relatively* more monosemantic, with superposition scores 1.03 standard deviations below that expert’s average.

We then test the reverse direction: when an expert represents features monosemantically, does it also occupy those feature directions in input space? We measure expert usage when those features are *one of many* active features and when they are the *only* active features; the latter directly estimates whether the expert occupies those directions. Table 1 shows that the correlation holds under both xavier and k-hot initialization. Since  $E = 10$ , mean usage near  $\sim 10\%$  indicates balanced use across experts. For every monosemantic-

feature group, expert usage rises substantially when the corresponding feature or features are active, and the expert dominates when those features are the only active inputs. Under k-hot initialization, 100% of features represented monosemantically by an expert are also occupied by that same expert.

Table 1: Feature-based specialization statistics for  $n = 100, m = 10, E = 10$ . For each initialization scheme (xavier and k-hot), we train one hundred models, yielding 1000 experts. Experts are grouped by how many features they represent monosemantically. For each expert’s monosemantic feature set, we report overall usage, usage when that feature set is active among other features, and usage when that feature set is the *only* active input.

| Xavier Initialization                      |                                 |                       |                                   |                                        |
|--------------------------------------------|---------------------------------|-----------------------|-----------------------------------|----------------------------------------|
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean usage; feature(s) active (%) | Mean usage; only feature(s) active (%) |
| 0                                          | 461                             | –                     | –                                 | –                                      |
| 1                                          | 387                             | 9.595                 | 17.94                             | 67.18                                  |
| 2                                          | 138                             | 9.599                 | 30.29                             | 95.65                                  |
| 3                                          | 13                              | 8.363                 | 40.19                             | 100.0                                  |
| 4                                          | 1                               | 1.428                 | 14.69                             | 100.0                                  |
| 5                                          | 0                               | –                     | –                                 | –                                      |
| K-Hot Initialization                       |                                 |                       |                                   |                                        |
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean usage; feature(s) active (%) | Mean usage; only feature(s) active (%) |
| 0                                          | 335                             | –                     | –                                 | –                                      |
| 1                                          | 382                             | 10.00                 | 23.94                             | 100.0                                  |
| 2                                          | 227                             | 10.02                 | 46.61                             | 100.0                                  |
| 3                                          | 47                              | 10.09                 | 62.00                             | 100.0                                  |
| 4                                          | 8                               | 9.95                  | 70.30                             | 100.0                                  |
| 5                                          | 1                               | 9.62                  | 74.79                             | 100.0                                  |

Table 1 supports the proposed specialization criterion. Experts that represent more features monosemantically behave more selectively: their unconditional usage is not the main signal, but their conditional usage rises sharply when their represented features are active. The pattern holds across the table except for the xavier-initialized case with one four-feature monosemantic expert, where utilization drops substantially.

## 7 CONCLUSION

This paper contributes a mechanistic account of how MoE routing changes superposition in toy reconstruction models. First, MoEs partition feature interference across experts and exhibit greater monosemanticity than dense networks with comparable budgets. Second, increasing network sparsity removes the sharp dense-model phase changes driven by feature sparsity and feature importance, replacing them with more continuous expert-specific transitions. Third, expert specialization is better captured by the alignment between routed feature directions and monosemantic feature representation than by load balance alone. These findings are limited to simple autoencoder toy models with synthetic data, so their transfer to large-scale transformers with unknown feature distributions remains open (Appendix A.6). Within that scope, the results show that MoEs can approach dense-model reconstruction loss while maintaining more interpretable representations, challenging the assumption that mechanistic interpretability and model capability must always trade off. Future work should identify what favors monosemanticity in MoEs, how MoE training dynamics differ from dense-model dynamics, and when feature-based specialization emerges at scale.

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## A APPENDIX

### A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

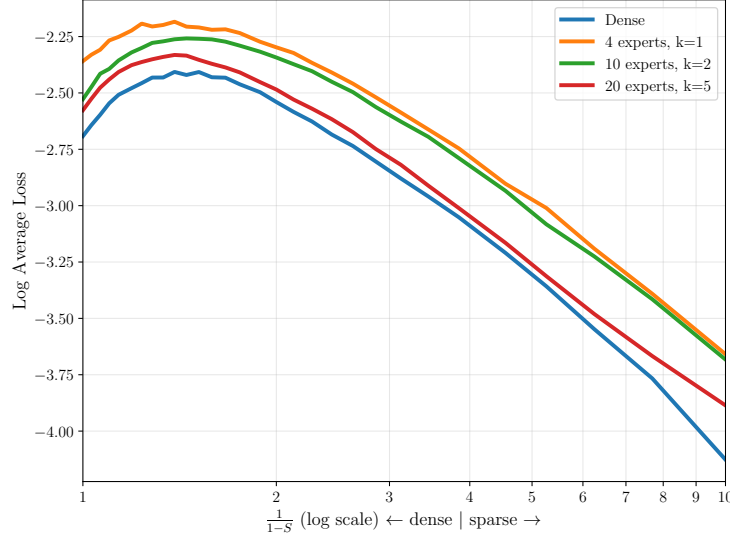


Figure 6: Reconstruction-loss comparison supporting the main superposition result. We plot log average loss versus feature density ( $\frac{1}{1-s}$ ) for **dense** ( $m = 20$ ) and **MoE (4 experts,  $k = 1$ ,  $m = 5$ )**, **MoE (10 experts,  $k = 2$ ,  $m = 2$ )**, and **MoE (20 experts,  $k = 5$ ,  $m = 1$ )** models, all with uniform feature importance ( $I_i = 1.0$ ) and  $n = 100$  input features. Results are averaged over five runs per sparsity level. The **dense** model has lower loss at every sparsity level, but MoE loss approaches the dense loss as expert count increases.

### A.2 THEORETICAL INTUITION FOR SUPERPOSITION

The intuition behind our main contribution is that dense models and MoEs exploit sparsity differently. Dense models use superposition to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs can instead “eat this same gap” structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs reduce the computational penalty for dedicated, rarely activating neurons. As Elhage et al. (2022) note, when computation is spent only on active features, splitting polysemantic neurons into monosemantic ones becomes preferable: MoEs trade the *compression* of superposition for the *selection* of routing.

Geometrically, this structural change appears as reduced *interference*. Even when the global ratio of represented features to parameters is held constant, the router partitions feature space so that a feature routed to expert  $e$  competes only with features assigned to that same expert. Interference is therefore governed by the local expert matrix  $(W^e)^\top W^e$  rather than the dense model’s global  $W^\top W$ . This partitioning lowers the number of possible interfering features for any given direction, reducing pressure to pack features in superposition and making monosemantic expert representations more likely.

### A.3 ROUTER SUBSPACES ARE CONVEX CONES

For  $k = 1$ , the router  $g(x) = \text{softmax}(W^r x)$  selects the same expert as  $\text{argmax}(W^r x)$ . The region routed to expert  $i$  is therefore described by  $\forall j \neq i, (w_i - w_j)^\top x > 0$ , where  $w_i$  and  $w_j$  are rows of  $W^r$ . These homogeneous linear inequalities define convex cones. If  $x$  satisfies the inequalities, then any positive rescaling  $sx$  also satisfies them; if  $x_1$  and  $x_2$  satisfy them, then every convex combination  $x = x_1\lambda + (1 - \lambda)x_2$  for  $\lambda \in [0, 1]$  satisfies them as well. This geometric fact supports our use of “occupancy” for feature directions.

For  $k > 1$ , the set of inputs sent to a particular expert  $e$  becomes a union of convex cones. Because a union of convex cones is not generally a convex cone, the clean occupancy interpretation used for the specialization contribution may not extend beyond  $k = 1$ .

#### A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

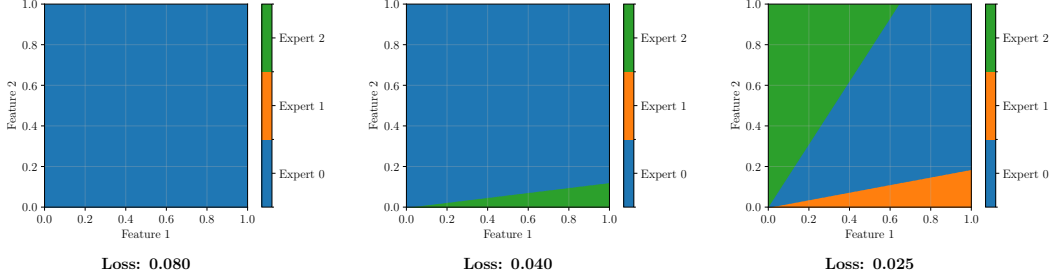


Figure 7: Routing specialization in three identical  $n = 2$ ,  $E = 3$ ,  $m = 1$  models with different initialization schemes. The left model has the highest loss (0.08) and routes all inputs to one expert. The middle model improves loss (0.04) by routing a small portion of inputs, specifically cases where feature 1 is active, to a second expert. The right model has the lowest loss (0.025) and partitions the input space across all experts: one for feature 1 alone, one for feature 2 alone, and one for both features active.

In small models ( $n = 2$ ,  $m = 1$ ,  $E > 1$ ), experiments with  $E \in [2, 7]$  show that lower loss tends to coincide with distributing input space across more experts. Holding  $n = 2$  makes this specialization geometry visible: Figure 7 shows which input regions route to which experts.

#### A.5 ANALYTIC MODEL EQUIVARIANCE

This appendix gives theoretical intuition for why the main text finds less superposition in top- $k = 1$  MoEs. In the single-layer, single-nonlinearity toy setup, any dense model can be mapped to a monosemantic MoE with an equivalent number of active features, provided a sparsity constraint holds.

Assume an upper bound  $a$  on the number of active features in any input, so  $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$ . Also assume that this active-feature bound is no larger than an expert’s hidden dimensionality  $m$ , and that  $m$  is no larger than the number of input features  $n$  ( $a \leq m \leq n$ ). To construct a monosemantic MoE, create an expert for each possible subset  $S \subseteq \{1, 2, \dots, n\}$  with  $|S| \leq a$ , where the subset size is at most the number of active features, and have that expert preserve those features monosemantically. (It is sufficient to take only subsets with  $|S| = a$ .) The router then selects the expert corresponding to the active feature set, which has size at most  $a$  by assumption:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where  $\text{support}(x) = \{i : x_i \neq 0\}$ . Because  $|S| \leq a \leq m$ , each expert has enough capacity to represent its assigned features without superposition. The construction therefore makes the contribution of routing explicit: at most  $a$  features are active, and every active-feature combination receives an expert with sufficient capacity for monosemantic representation. The number of possible experts required is  $\binom{n}{a}$ .

This theoretical MoE achieves zero reconstruction loss only when the sparsity constraint holds, or in these toy models as sparsity approaches one. If more than  $m$  features can be active at once ( $a \not\leq m$ ), the active set exceeds the monosemantic representational capacity of an expert; a dense polysemantic model faces the same capacity issue unless the data distribution contains helpful feature correlations. Thus, even when  $a \not\leq m$  occasionally,

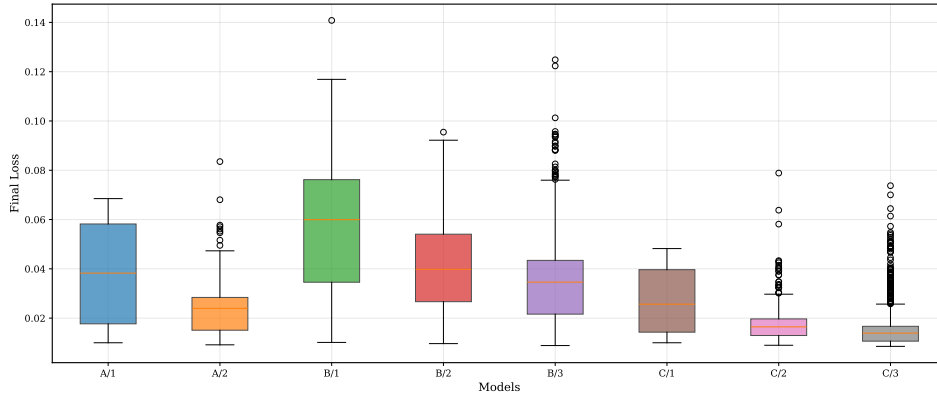


Figure 8: Loss patterns for the same architectures used in Figure 4. In model label  $X/E$ ,  $X$  denotes the architecture and  $E$  denotes the number of experts, i.e. network sparsity. Increasing network sparsity lowers mean loss while increasing localized variance, especially when expert count reaches the number of input feature dimensions. This variance is consistent with less stable MoE training relative to dense models, even though ten models are trained per cell and the lowest-loss run is selected.

the constructed monosemantic MoE may still outperform a polysemantic alternative under looser sparsity constraints.

Therefore, for any dense model  $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$  satisfying the sparsity constraint  $|\text{support}(x)| \leq a$ , there exists a MoE model  $f_{\text{MoE}}(x)$  such that  $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$  for all valid inputs. In the toy settings studied here, the constraint holds in the limit as sparsity approaches one. In practice, meaningful data may impose an upper bound on how many features a fixed amount of information can semantically encode, as reflected by embedding size. Under such conditions, a MoE with enough experts and tractable residual superposition may be sufficient to encode all present features while remaining more interpretable.

## A.6 LIMITATIONS

The paper’s contributions should be read within several limitations. First, every experiment uses controlled toy models derived from sparse-autoencoder superposition studies (Elhage et al., 2022). This design is what lets us measure interference, monosemanticity, and expert occupancy precisely, but it removes many features of large-scale transformers, including multi-layer computation, attention, heterogeneous feature distributions, and realistic routing dynamics. As a result, transfer to deployed MoE architectures remains an open question.

Second, the architecture is deliberately restricted to make the contribution attributable to network sparsity: we use simple experts, fixed top- $k$  routing (often  $k = 1$ ), matched dense/MoE parameter budgets, and omit auxiliary load-balancing losses except where stated. Practical MoEs often use richer routers, variable expert capacities, multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), and objectives beyond reconstruction, such as next-token prediction. Those choices may produce qualitatively different representational behavior.

Third, the synthetic feature distributions are much simpler than natural data. Their sparsity patterns, feature-importance schedules, and independence assumptions make the contribution clean to measure but do not capture the structure of real language or multimodal inputs.

Finally, our evidence for increased monosemanticity and reduced superposition is restricted to fixed toy conditions. We do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale, and prior work suggests that these dynamics can change with model size, optimization regime, and data diversity (Krajewski et al., 2024).

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810 Overall, the study contributes mechanistic evidence under clean experimental conditions  
811 rather than a claim that the same patterns necessarily hold in full-scale MoE systems.  
812 Validating network sparsity, phase-change behavior, and feature-based specialization in re-  
813 alistic models will require methods that scale beyond the measurements used here, which is  
814 a central direction for future work.  
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